

CS5275 – The Algorithm Designer's Toolkit
(S2 AY2025/26)

Lecture 12:

Boolean functions – social choice functions

Social choice functions / voting rules

A Boolean function

$$f : \{-1,1\}^n \rightarrow \{-1,1\}$$

can be viewed as a **voting rule** for an election with 2 candidates and n voters.

also known as social choice function



$\{-1,1\}$



The input $x \in \{-1,1\}^n$
represents their votes.

Common voting rules

$$\text{Majority: } \text{Maj}_n(x) = \text{sgn}(x_1 + x_2 + \cdots + x_n)$$

$$\text{sgn}(s) = \begin{cases} 1 & s > 0 \\ -1 & s < 0 \\ \text{can be 1 or } -1 & s = 0 \end{cases}$$

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also known as **linear threshold function**

$$a_0, a_1, \dots, a_n \in \mathbb{R}$$

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Special cases of
weighted majority

$$\textbf{The } i\text{th dictator function: } \chi_i(x) = x_i$$

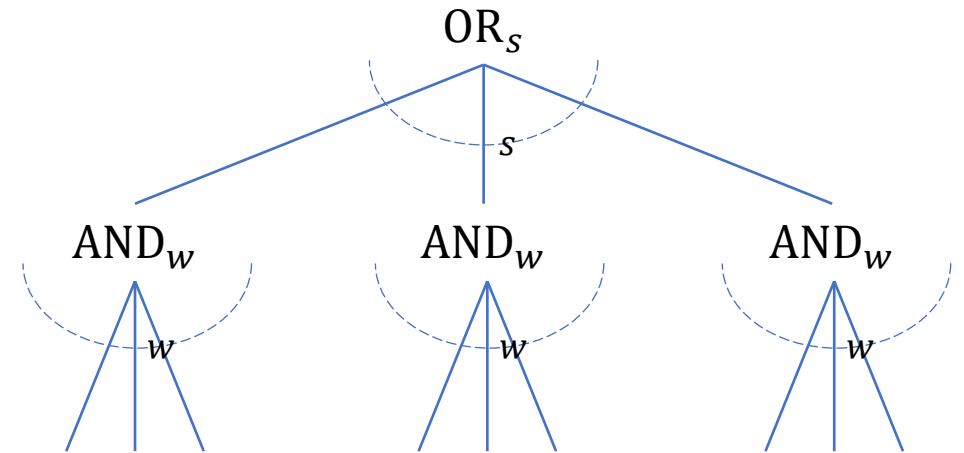
$$\textbf{AND: } \text{AND}_n(x) = \begin{cases} -1 & x = (-1, \dots, -1) \\ 1 & \text{otherwise} \end{cases}$$

$$\textbf{OR: } \text{OR}_n(x) = \begin{cases} 1 & x = (1, \dots, 1) \\ -1 & \text{otherwise} \end{cases}$$

$$\begin{aligned} -1 &= \text{true} \\ 1 &= \text{false} \end{aligned}$$

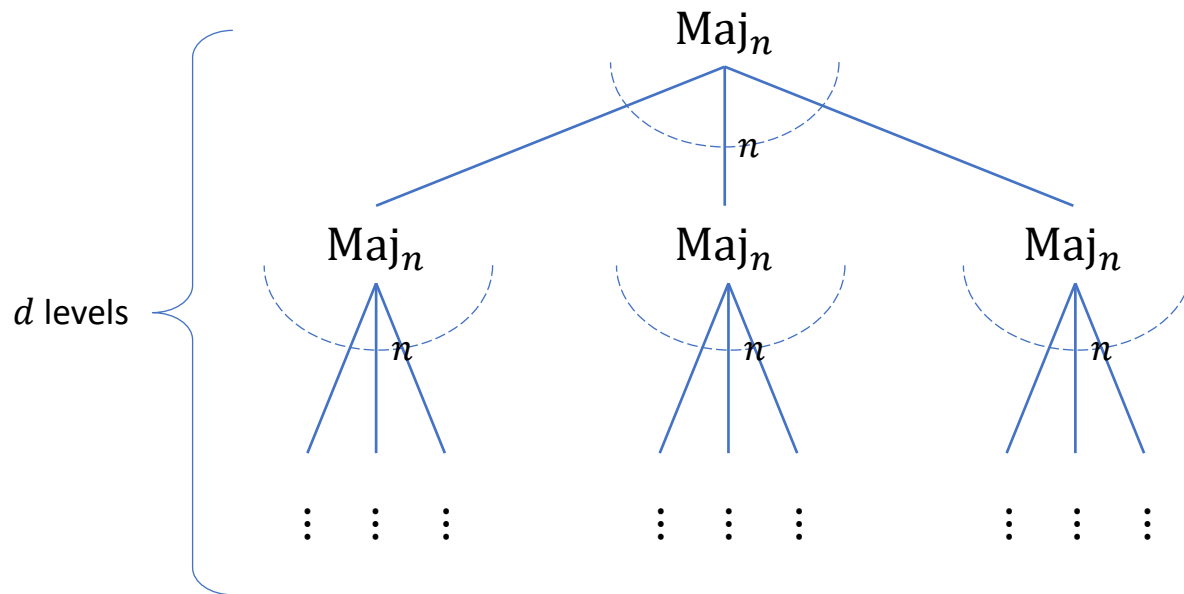
Common voting rules

Tribes: $\text{Tribes}_{w,s} : \{-1,1\}^{sw} \rightarrow \{-1,1\}$

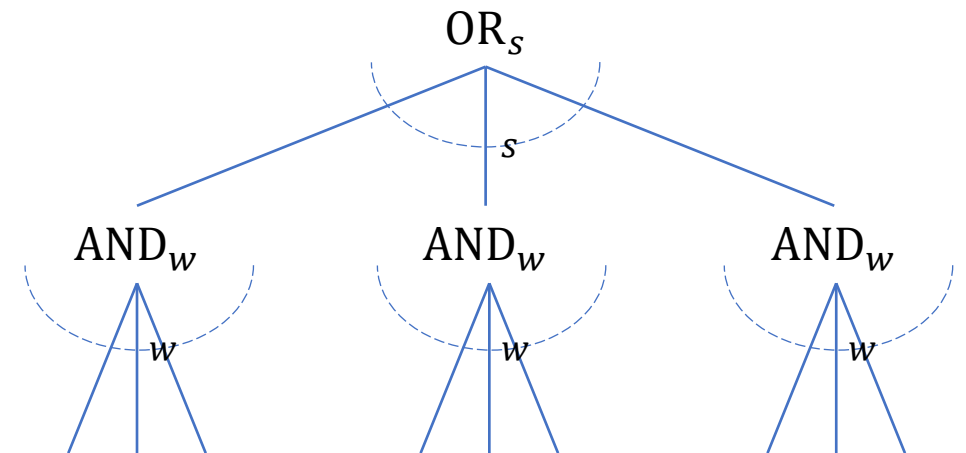


Common voting rules

Depth- d majority: $\text{Maj}_n^{\otimes d} : \{-1,1\}^{n^d} \rightarrow \{-1,1\}$



Tribes: $\text{Tribes}_{w,s} : \{-1,1\}^{sw} \rightarrow \{-1,1\}$



Desirable properties

Monotone: $f(x) \leq f(y)$ whenever $x_i \leq y_i$ for all i

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Monotone	✓	✓	✓	✓
Odd	✓	✓	✗	✗
Unanimous	✓	✓	✓	✓
Symmetric	✓	✗	✓	✗

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Exercise

May's theorem: Maj_n is the only monotone, odd, symmetric voting rule.

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$\text{Tribes}_{w,s}$ is **transitive-symmetric**:

- for all $i, i' \in [n]$, there is a permutation $\pi \in S_n$ with $\pi(i) = i'$ such that:
 - for all $x \in \{-1, 1\}^n$, $f(x^\pi) = f(x)$

Influence of a voter

$$x^{\oplus i} = (x_1, \dots, x_{i-1}, -x_i, x_{i+1}, \dots, x_n)$$

$i \in [n]$ is **pivotal** for $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ on input x if $f(x) \neq f(x^{\oplus i})$

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The **influence** of a coordinate i on f is $\text{Inf}_i[f] = \Pr_{x \sim \{-1, 1\}^n}[i \text{ is pivotal for } f] = \mathbb{E}_{x \sim \{-1, 1\}^n}[f(x) \neq f(x^{\oplus i})]$

Impartial cultural assumption: $x \sim \{-1, 1\}^n$

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Geometric interpretation:

$\text{Inf}_i[f]$ is the fraction of the dimension- i edges that cross the cut represented by f in the n -dimensional hypercube.

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$$\text{Inf}_i[\chi_j] = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$\text{Inf}_i[\text{AND}_n] = \text{Inf}_i[\text{OR}_n] = 2^{1-n}$$

$$\text{Inf}_i[\text{Maj}_n] \in \sqrt{\frac{2}{\pi n}} + O(n^{-1.5})$$

Exercises

$\text{Inf}_i[f] \leq 1/\sqrt{n}$ if f is **transitive-symmetric** and **monotone**.  **Next**

Influence of a voter

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$$x^{i \mapsto b} = (x_1, \dots, x_{i-1}, b, x_{i+1}, \dots, x_n)$$

Derivative operator:

$$D_i f(x) = \frac{f(x^{i \mapsto 1}) - f(x^{i \mapsto -1})}{2}$$

↓

$$\text{Inf}_i[f] = \mathbb{E}_{x \sim \{-1, 1\}^n} [(D_i f(x))^2] = \|D_i f(x)\|_2^2$$

The generalization of influence to any real-valued function $f : \{-1, 1\}^n \rightarrow \mathbb{R}$

Influence of a voter

$$\text{Claim: } \text{Inf}_i[f] = \sum_{S \ni i} \hat{f}(S)^2$$

Parseval's theorem

$$D_i \chi_S = \begin{cases} \chi_{S \setminus \{i\}} & i \in S \\ 0 & i \notin S \end{cases} \longrightarrow D_i f(x) = \sum_{S \ni i} \hat{f}(S) \chi_{S \setminus \{i\}}$$

$$x^{i \mapsto b} = (x_1, \dots, x_{i-1}, \textcolor{red}{b}, x_{i+1}, \dots, x_n)$$

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$f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is **monotone**

$$\begin{aligned} \text{Inf}_i[f] &= \mathbb{E}_{x \sim \{-1, 1\}^n} [D_i f(x)] \\ &= \widehat{D_i f}(\emptyset) \\ &= \hat{f}(\{i\}) \end{aligned}$$

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$$\text{Inf}_i[f] \leq 1/\sqrt{n}$$

$$x^{i \mapsto b} = (x_1, \dots, x_{i-1}, \textcolor{red}{b}, x_{i+1}, \dots, x_n)$$

$$1 = \sum_{S \subseteq [n]} \hat{f}(S)^2 \geq \sum_{i \in [n]} \hat{f}(\{i\})^2 = n \cdot (\text{Inf}_i[f])^2$$

f is **transitive-symmetric**

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The generalization of influence to any real-valued function $f : \{-1, 1\}^n \rightarrow \mathbb{R}$

Total influence

Total influence: $I[f] = \sum_{i \in [n]} \text{Inf}_i[f]$

Geometric interpretation:

$\frac{I[f]}{n}$ is the fraction of the edges that cross the cut represented by f in the n -dimensional hypercube.

Total influence

Total influence: $I[f] = \sum_{i \in [n]} \text{Inf}_i[f]$

Suppose $f : \{-1,1\}^n \rightarrow \{-1,1\}$ is **monotone**.

$$\text{Inf}_i[f] = \hat{f}(\{i\})$$

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Observation: Under the Impartial cultural assumption, the expected number of votes that agree with the outcome of the election is

$$\frac{n}{2} + \frac{1}{2} \cdot \sum_{i \in [n]} \hat{f}(\{i\})$$

$$\sum_{i \in [n]} \hat{f}(\{i\}) = \sum_{i \in [n]} \mathbb{E}_{x \sim \{-1, 1\}^n} [f(x) x_i] = \mathbb{E}_{x \sim \{-1, 1\}^n} [f(x) (x_1 + \cdots + x_n)]$$

(votes for winner) — # (votes for loser)

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Corollaries:

- Maj_n is the unique maximizer of $\sum_{i \in [n]} \hat{f}(\{i\})$ over all functions $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$.
- $I[f] \leq I[\text{Maj}_n]$ for any **monotone** $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$.

Total influence

Total influence: $I[f] = \sum_{i \in [n]} \text{Inf}_i[f]$

$$\text{Inf}_i[f] = \sum_{S \ni i} \hat{f}(S)^2 \longrightarrow I[f] = \sum_{S \subseteq [n]} |S| \hat{f}(S)^2$$

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Poincaré inequality

Exercise: The conductance of n -vertex hypercube is $\Omega(1/\log n)$.

Outlook

- For 2-candidate voting, it is quite clear that **majority** is the right choice.
 - May's theorem.
 - $I[f] \leq I[\text{Maj}_n]$ for any monotone $f : \{-1,1\}^n \rightarrow \{-1,1\}$.
- Things become more complicated when we have more than 2 candidates.
- **Next:** Arrow's impossibility theorem.

When there are more than 2 candidates, no voting system can convert individual preferences into a collective decision while satisfying a set of seemingly reasonable fairness conditions simultaneously.

References

- Sections 2.1–2.3 of <https://arxiv.org/abs/2105.10386>